

Diagnosing and Surgically Correcting Conservative Decision Bias in Imitation-Learned Air Combat Policies

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ABSTRACT

Behaviour cloning (BC) from rule-based experts is a standard bootstrap for air-combat policies, under the implicit assumption that the expert's decisions are worth copying. On a JSBSim F-16 within-visual-range 1v1 missile-engagement benchmark we test that assumption and find it fails in a specific, measurable way: the hand-designed expert launches on only 6% of the steps its environment permits, and BC reproduces that restraint faithfully — a restrictive launch preference encoded in the expert rule, not an imitation defect. We introduce the **Conditional Launch Rate (CLR)** to quantify it, and **Surgical Policy Correction (SPC)**, a localized intervention that retrains *only* the launch head while every other parameter — and therefore the entire maneuver policy — stays bit-identical. Correcting that single decision dimension recovers the performance of an oracle that fires whenever the environment permits launch: 46.50% → 90.75% kill rate on 400 paired seeds (+44.25 pp, exact McNemar $p = 1.04e-53$, all 177 discordant seeds favouring the correction), matching the level a frozen-trajectory intervention reaches in the hardest close-range cell (15.0% → 86.7%). Under an evading target the benefit grows rather than decays (+48.50 pp). The measured failure mode was localized to one discrete decision dimension, and correcting only that dimension recovered oracle-level performance while preserving the original maneuver — the oracle being a reference defined by the environment's launch mask, not a claimed optimum.

KEYWORDS

imitation learning, behaviour cloning, expert-induced bias, causal identification, policy intervention, air combat

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1 INTRODUCTION (1 PAGE)

1.1 Background

- Imitation learning (IL) bootstraps control policies in domains where online trial-and-error is expensive or unsafe: autonomous driving, manipulation, air combat.
- BC is the simplest IL: supervised regression/classification on expert demonstrations.
- Dominant assumption: the expert is a *sufficient* teacher; fidelity to the expert is the objective.

1.2 Problem: expert-induced bias, amplified by closed-loop interaction

- Rule/heuristic experts are designed, not optimized. Their decision rules encode engineering judgement that is *locally* plausible but never validated in the closed loop.
- In interactive settings (opponent/evader present), a locally reasonable restriction — "only launch when the shot is high quality" — changes the entire engagement trajectory, so its cost is not locally observable.
- BC propagates the bias faithfully, so downstream RL or evaluation inherits it as a *floor*, not a *fluctuation*.
- Existing BC evaluation reports aggregate task success only; it does not attribute the residual to a specific decision head.
- **Terminology, fixed once (2026-09-18):** we say **restrictive launch preference** when describing the *mechanism* — the expert's designer would call that gate *precision*, and locally it is; we reserve **expert-induced decision bias** for the *measured, globally costly consequence* of inheriting it. We do not assume the expert rule is erroneous in isolation; we test whether its preference remains beneficial under closed-loop interaction, which is a different question.

1.3 Contributions

- (1) **C1 — Diagnosis: expert-induced decision bias, quantified.** We introduce the **Conditional Launch Rate (CLR)**, $P(a_{\text{fire}}=1 \mid m_{\text{fire}}=1)$ — an oracle-free metric that transfers to any commit-or-not decision (launch, hand-off, emergency stop, lane change) — and find that the rule expert launches on only 6% of the steps its environment permits, with BC reproducing that preference faithfully (7%). We localise the cause to a **designed** launch-quality gate that is strictly stricter than the environment's legality mask (\S3.2). This is not a case of the expert being incompetent: the expert carries a **restrictive launch preference the environment does not require**, and its closed-loop cost is the decision bias this paper measures.
- (2) **C2 — Surgical Policy Correction: a localized *intervention* on the decision dimension the diagnosis identifies.** SPC modifies **only the single binary launch head** so that it fires whenever the environment permits, while **every other parameter — and therefore the entire maneuver policy — is provably preserved** (bit-identical heading/speed logits and action sequences, \S3.3). The contribution is not the implementation detail of training one head; it is *isolating the behavioural dimension responsible for the failure*: the maneuver is bit-identical, so the residual difference can only be attributed to the launch decision. A5/A6 (\S4.7) then show that the benefit comes from **which parameters**

are allowed to move — not from how many, and not from where that head started.

- (3) **C3 — Causal and locality validation: two interventions, one conclusion.** Two complementary causal arguments (\S4.3, \S4.4), plus the A5/A6 validation that locality is the key factor (\S4.7). **The two interventions live at different scales, and each must be cited with its own scope attached:**
- **Intervention A (cell-specific frozen-trajectory enumeration, dist2_3k, 60 seeds):** 15.0% → 86.7% — *the launch decision alone can explain the performance gap*;
 - **Intervention B (primary 400-seed policy intervention, seeds 20000–20399):** 46.50% → 90.75% (paired, 177:0, $p \approx 1e-53$) — *the correction can be internalized into the policy*.
 - The effect survives and grows under an evading target (+44.25 → +48.50 pp), and is replicated by an independent identification strategy (\S4.3 E6 matched 2×2: +40.0 → +50.0 pp).

Structure of the argument (the body follows this, one level above a conventional ablation report):

2 RELATED WORK (1 PAGE)

2.1 Imitation learning and expert demonstration

- **BC and its standard pathology.** Cloning a demonstrated policy by supervised learning goes back to ALVINN (Pomerleau, NIPS 1:305-313, 1989 [verify year]) and was scaled to end-to-end driving by Bojarski et al. (arXiv:1604.07316, 2016). The canonical account of its failure mode is DAgger (Ross, Gordon & Bagnell, AISTATS 2011), which attributes the failure to **distribution shift / compounding error** between the learner's and the expert's state visitation.
- **Core sentence.** *Existing IL methods usually assume the demonstrations are informative and near-optimal; the learner's objective is fidelity to the expert.*
- **The literature that does relax optimality relaxes it as a scalar.** T-REX learns a reward from ranked suboptimal demonstrations (Brown, Goo, Nagarajan & Niekum, ICML 2019, pp. 783-792); LERP models suboptimality as reward noise (Huo, Wang & Xu, AAAI 2023, pp. 7953-7961); IRLEED models a demonstrator's suboptimality as *reward bias plus action variance* (Beliaev & Pedarsani, arXiv:2402.01886, 2024); IL-from-imperfection reweights supplementary suboptimal data (Li, Xu, Qin, Yu & Luo, NeurIPS 2023). In every case suboptimality is a **scalar quality** to be estimated, ranked, or down-weighted.
- **Our gap.** *What if the expert's suboptimality is not scalar but structural - a single, nameable decision gate that the environment does not require?* A scalar quality cannot recover it: the defect is invisible in task-level success statistics (\S4.2) and costly only in closed loop. We do not down-weight the expert; we **localize** the gate and correct exactly it.

2.2 Policy distillation and model editing

- Knowledge distillation into a student (Hinton, Vinyals & Dean, NIPS 2014 Deep Learning Workshop, arXiv:1503.02531),

policy distillation (Rusu, Gomez Colmenarejo, Gulcehre et al., ICLR 2016, arXiv:1511.06295), and Actor-Mimic (Parisotto, Ba & Salakhutdinov, ICLR 2016, arXiv:1511.06342) all **transfer or compress a policy globally**, with fidelity to the teacher as the objective.

- Parameter-efficient adaptation (LoRA; Hu, Shen, Wallis, Allen-Zhu, Li, Wang, Wang & Chen, ICLR 2022, arXiv:2106.09685) and model editing (ROME; Meng, Bau, Andonian & Belinkov, NeurIPS 2022, arXiv:2202.05262, pp. 17359-17372) do modify a small subspace while freezing the remainder - but they edit **learned knowledge** and evaluate by task success, not by attributing a residual to one behavioural dimension.
- **Core sentence.** *Existing approaches transfer or modify policies globally; we study localized correction of a single decision dimension, and we use the freezing itself as a causal identification device.* The distinction from distillation is precise: SPC's objective is **not** to approach the teacher but to **selectively depart** from it on one head. Freezing is not an engineering convenience here - it is the identification strategy (\S4.4).

2.3 Autonomous air combat decision-making

- **Closest neighbours, stated plainly.** Li et al. (Neurocomputing 584:127591, 2024) learn a within-visual-range 6-DOF combat policy whose top layer decides autopilot commands **and missile launch**, with BC and PPO cross-coordinated. Li et al. (ACM TAAS 21(1):6:1-6:20, 2026) learn a "pursuit-lock-launch" policy with TD3 + BC and an adaptive imitation weight in Harfang3D. Both clone a launch decision from demonstrations in a WVR missile-engagement setting.
- **Where we differ - two ways, both load-bearing.** (i) They treat the expert as a **bootstrap to be surpassed**; we treat it as an **object of diagnosis**, and we keep the maneuver provably frozen so the residual is attributed to the launch head rather than to aggregate learning. (ii) Their success criterion is aggregate task success; we add a **conditional** criterion (CLR) that exposes failures aggregate success hides. Notably, Li et al. (2026) present selective firing as a virtue - their policy "fires only when a hit was highly probable rather than spamming launches" - the **opposite prior** to the one this paper tests, and in our regime the measurement goes the other way (\S4.3).
- Background only, kept short: approximate dynamic programming for air combat (McGrew, How, Williams & Roy, JGCD 33(5):1641-1654, 2010); hierarchical maximum-entropy RL for an F-16 in within-visual-range combat (Pope et al., ICUAS 2021, pp. 275-284, arXiv:2105.00990); short-range UAV maneuver decisions via deep RL (Yang, Zhang, Shi, Hu & Wu, IEEE Access 8:363-378, 2020). JSBSim is the open-source flight-dynamics model underneath all of these.
- **Constraint.** This subsection stays under half a page. It is scene-setting, not contribution positioning: a reviewer should enter this paper through \S2.1/\S2.2, not file it as another air-combat RL paper.

3 METHOD (1.5 PAGES)

3.1 Task and policy

- **Task:** 1v1 **within-visual-range** air combat, JSBSim F-16 dynamics; pursuer carries 4 short-range IR missiles (AIM-9L class); initial range 2–5 km tail-chase, target heading bias 0–60° (default), target speed U(180,240) m/s. Max engagement range 15 km.
- **Observation:** 41-dim flattened = 30-dim state features + 11-dim action-mask bits. The BC encoder consumes the 30-dim state block; the mask block is applied as a separate masked-argmax step, not fed to the network.
- **Action space:** MultiDiscrete([3, 5, 1, 2]) = (speed Δ , heading Δ , altitude, fire).
- **Episode:** max 1500 steps; termination = target killed / low altitude / fled altitude / target lost / ammo exhausted / time-out.
- **Step duration — state it explicitly, it is a units trap.** One environment step advances the simulation by **0.2 s** (BaseEnv._acmi_time += 0.2), i.e. the agent runs at **5 Hz**; 1500 steps $\approx$ 300 s. This is *not* the JSBSim internal frame rate. Every step $\rightarrow$ second conversion in this paper uses 0.2 s/step (\S4.3: 67 steps = 13.4 s, 743.5 steps = 149 s). Any figure or table that appears to use 1/60 s per step is wrong by 12 $\times$.
- **Evaluation protocol:** **deterministic masked argmax**, JSBSim physics, difficulty_level as noted per experiment.
- **Three policies in play:**
- **Rule expert** — hand-designed, stateless: every label is a pure function of the current observation (scripts/generate_shoot_rule_expert.py). **Provenance, stated plainly: there is no external publication to cite.** It is this project's own rule controller, built from standard BFM heuristics (nose alignment, DLZ depth, closure sign) plus the launch-quality gate of \S3.2, and trained against nothing. We do not claim it is a strong or a state-of-the-art expert; we claim only that it is a competent teacher (wez_reach_rate = 1.00, hit_rate $\approx$ 0.997, lost_target_rate = 0) whose behaviour carries a measurable conservative bias. Every headline claim in this paper is about *what imitation inherits from a given teacher*, holding the teacher fixed — so the paper does not depend on where the teacher came from, and inventing a citation for it would be worse than saying this.
- **BC round1** — MLP encoder (256-256-128) + 4 heads, trained on 200 expert episodes, masked cross-entropy.
- **SPC** — BC with only the launch head re-trained (\S3.3).

3.2 C1 — Diagnostic metric and localization

Conditional Launch Rate (CLR):

- m_fire is the **environment's** legality mask (index 10 of the flat mask), which requires: missiles remaining > 0; launch cooldown (30 decision steps) elapsed; target alive; ATA < 15°; MIN_ATTACK_DISTANCE (1500 m) $\leq$ range $\leq$ dynamic max range (3–8 km by aspect angle); closure < 0 (closing).

- **Critical property: CLR is not a claim that 100% is optimal.** It is a descriptive quantity. The optimal value must be established separately — which is exactly what \S4.3 does.

Why the expert's CLR is low — the designed gate. The expert does not simply follow m_fire . It requires its own stricter quality predicate:

Measured — the gate fully determines the expert's behaviour on legal steps.

$\Rightarrow$ Under the evaluated protocol, **the observed restriction is explained by the designed gate** — not by the environment and not by BC's fitting error.

Two further measurements corroborate the magnitude on a different basis — expert CLR 5.96% (560/9395) read directly off the demonstration dataset, against BC 7.37% (614/8327) from a 200-rollout diagnostic. They are quoted as corroboration only, and one caveat travels with them: the dataset predates the widening of the heading-bias range, so its 5.96% is an old-geometry measurement while the 7.37% beside it is not. They are never averaged, never presented as a matched pair, and the authoritative figure remains the re-measured 6.07%. Appendix E.2 gives the timestamps and the gate's behaviour on disallowed steps.

3.3 C2 — Surgical Policy Correction (SPC)

- **Input:** frozen BC θ_BC ; BC rollouts collected over R episodes.
- **Target:** on steps where $m_fire = 1$, command fire. All other steps are irrelevant (the mask forces $a_fire = 0$ at inference).
- **Procedure:**
 - (1) Roll out θ_BC ; collect (obs_t, m_fire , t).
 - (2) Filter to $m_fire, t = 1$.
 - (3) Train **only** action_heads[3] by cross-entropy toward $a_fire = 1$. Encoder, speed head, heading head, altitude head: requires_grad_(False).
 - (4) Early stop when allowed-window fire accuracy ≥ 0.995 .
- **Verification gates** (all must PASS; implemented in scripts/distill_fire_asap.py):
 - G1 heading/speed logits vs θ_BC : max diff < 1e-9
 - G2 heading/speed **action sequences** 100% identical on fixed seeds
 - G3 CLR $\geq 99.5\%$ over allowed windows ($P(a_fire=1 \mid m_fire=1) \geq 0.995$) on held-out rollouts
 - G4 no new lost_target / bad launch introduced
- **Interpretation:** G1–G2 are the causal-identification device — the maneuver policy is provably untouched, so any performance change is attributable to the launch decision alone.

3.4 Why isolating the launch head is the right intervention (not arbitrary)

- The launch decision is a **discrete binary commitment** with a one-step-wide opportunity in the close-range regime (\S4.3), so its error is not smoothable by the rest of the policy.
- The maneuver heads are demonstrably near-ceiling: wez_reach_rate = 1.0, hit_rate $\approx$ 0.997–1.000, lost_target_rate = 0 across all evaluated suites. Re-training them would buy little and destroy the clean attribution.

Condition	Environment mask	Expert fire_desired	Effect
Nose alignment	ATA < 15°	ATA < 10°	refuses 10–15° shots
Closure	closure < 0	closure < -5 m/s	refuses marginal closure
DLZ position	legality only	depth ∈ [0.25, 0.75]	refuses deep-in-DLZ (closer) and far-edge shots
Output	—	fire = allowed AND desired	cooldown/inventory handled by mask

Quantity	Value	Basis
Expert CLR (fresh env, 400 ep, seeds 20000–20399, U(0,60))	6.07% (1131 / 18639)	E3_paired_bc_vs_expert_d0_n400_s20000_v3_clr.json
Expert fire_desired on allowed steps	560 / 9395 — identical to action[:, 3] ⇒ nothing except the gate explains what the expert does when launch is legal	data/expert/shoot_rule_expert.npz

Policy	CLR (d=0)	CLR (d=0.3)	Legal steps (d=0 / d=0.3)
Rule expert	6.07% (1131/18639)	5.40% (1066/19729)	18639 / 19729
BC round1	7.26% (1223/16839)	6.95% (1194/17169)	16839 / 17169
SPC	100.00% (1560/1560)	99.11% (1563/1577)	1560 / 1577

- Contrast with end-to-end PPO fine-tuning: the project’s own record (docs/summary_phase1.md \S4.5) shows that once SPC reproduces the legality policy, the current reward/environment by construction admits no launch-timing trade-off, so PPO adds no increment and carries regression risk. → report as a *documented negative result*, not as a claim about PPO in general.

4 EXPERIMENTS (2 PAGES)

4.1 Setup

- JSBSim F-16 physics; 1v1 within-visual-range missile engagement; **deterministic masked argmax**.
- One seed protocol throughout: **seeds 20000–20399, fresh environment per episode, geometry U(0,60)**. Hardware, wall-clock and reproduction commands: Appendix D/E.
- Three evaluation settings: **ID** (2–5 km tail-chase), **dist2_3k** (hard close-range cell), **target_evasive** (difficulty_level = 0.3).
- **All reported numbers are verified against version-controlled artifacts**; the per-number provenance record is Appendix E.

4.2 Diagnosis: CLR reveals a restrictive launch preference

The expert fires on only 6.07% of the steps its environment permits (1131/18639), and BC inherits the preference (7.26%). Under an evading target both barely move (**5.40%** / **6.95%**). The preference is a property of the expert’s designed launch gate — strictly stricter than the environment’s legality condition (\S3.2) — and imitation transmits it faithfully; it is not an imitation defect.

Table 2 — Conditional Launch Rate by policy and difficulty (fresh env, seeds 20000–20399, U(0,60), deterministic argmax):

Two caveats travel with this table. (a) SPC’s 100.00% is true *by construction* — its training target is “fire on every legal step” and gate G3 tests exactly that; it is not a finding. (b) The CLR denominator is policy-dependent: firing starts a 30-step cooldown that *removes* legality, so a policy that fires collapses its own opportunity set (SPC ≈ 3.9 legal steps/episode, BC ≈ 42). Cross-policy CLR is therefore an ordering of tendencies, not a like-for-like exploitation

fraction — which is why the causal claim rests on \S4.3’s frozen-trajectory enumeration, where the opportunity set is held constant by construction.

One cross-check on the instrument itself: adding CLR changed nothing measurable. The d=0 run that first reported CLR reproduces the kill side of the E3 pair bit-for-bit (expert 36.75%, BC 46.50%, paired +9.75 pp, W/L/T 63/24/313) and reproduces BC’s CLR to the digit, and the d=0.3 run does the same (25.75%, 42.25%, +16.50 pp, 85/19/296). Expert and BC also agree closely on every basis measured — 6.07% vs 7.26%, 5.40% vs 6.95%, and 5.96% vs 7.37% on the older geometry-mixed diagnostics — so the restriction is a stable property rather than sampling noise, and BC is in no meaningful sense more restrictive than its teacher. Appendix E.3 states the scope limit on what CLR alone can establish.

4.3 C3(a) — Off-policy enumeration on a frozen trajectory (the key identification)

Design. Take BC’s heading/speed decisions as a *frozen* maneuver trajectory per seed, and enumerate a family of fire policies on top of it. Because the maneuver is fixed by construction, the maneuver policy is controlled, and differences are attributable to the launch decision.

Table 3 — frozen-trajectory enumeration (cell dist2_3k). 60 seeds, geometry U(0,60), deterministic masked argmax, maneuver frozen per seed from the BC round-1 checkpoint (E1, regenerated 2026-09-17):

The italic column is the earlier 2026-08-06 audit, executed on the pre-widening geometry U(30, 60); the measured column is the current default U(0, 60). That two of the seven arms reproduce exactly (delay_60 10.0%, dlz_mid 0.0%) while asap moves from 73.3% to 86.7% indicates that the *script*’s behaviour is unchanged and that the benchmark itself became easier by ≈ +4.5 pp (\S4.5). Only the measured column is cited, always with its geometry attached.

Reachability audit (asap arm, measured): median first-legal step 67 (13.4 s); **cumulative legal-window length ≈ 4 steps per episode** (median 4.0) — i.e. roughly one legal step per DLZ transit;

Fire policy	Kill rate (measured)	docs, old geom U(30,60)	Launches/ep	Reach-4-launch
asap — fire whenever the launch mask permits	86.7%	73.3%	3.85	86.7%
delay_30 (first legal + 30)	38.3%	28.3%	3.25	38.3%
delay_60	10.0%	10.0%	2.78	10.0%
dlz_mid (depth 0.4–0.6)	0.0%	0.0%	1.27	0.0%
dlz_deep (depth ≥ 0.6)	0.0%	3.3%	1.12	0.0%
interval_100	0.0%	0.0%	1.95	0.0%
frozen BC (inherited)	15.0%	6.7%	2.37	15.0%

difficulty	fire policy	Kill	Launches/ep	Legal-window median
d=0	asap	93.3%	3.92	4.0
d=0	frozen BC (inherited)	53.3%	3.18	42.0
d=0.3	asap	96.7%	3.97	4.0
d=0.3	frozen BC (inherited)	46.7%	3.05	42.0
asap - inherited gap		+40.0 pp $\rightarrow$ +50.0 pp		

median step-1 $\rightarrow$ step-4 launch **743.5** (149 s); 52/60 killed, 8 time-outs, **8 of which ended with the window still open** (blockers = episode_end_in_window: 8). The " ≈ 4 legal steps per episode" figure is the load-bearing structural fact and it reproduces the 2026-08-06 audit.

Reading:

- The scenario is **not infeasible** — the environment permits a 4-launch salvo in 86.7% of episodes, and asap realizes 86.7% kills.
- The bottleneck is the launch policy: the inherited policy uses **2.37 of 3.85** available windows, because most windows have DLZ depth < 0.25 and are rejected by fire_desired.
- **Every** selective alternative is worse than the mask-permissive policy, and by a wide margin (best rival delay_30 at 38.3%). Within the enumerated family and on this trajectory class, the mask-permissive policy **achieves the highest kill rate**; the comparison is confined to that family and is not a claim about fire policies in general.
- Note what this does and does not show: it establishes that the inherited launch policy is **dominated within the enumerated family, conditional on the frozen maneuver**. It does not establish that the mask-permissive policy is optimal for arbitrary maneuvers of arbitrary policies. (\S4.4 supplies the within-policy counterpart.)
- **Terminology discipline.** The mask-permissive policy is *not* "always fire": it fires only on steps where the environment-level launch mask is 1. The correction removes a gate the expert added *on top of* the environment's legality condition; it does not relax the environment's condition itself.

Same enumeration under an evading target — matched 2 $\times$ 2 (E6, seeds 20000–20059, n=60, geometry U(0,60), deterministic masked argmax).

target_evasive changes difficulty *and* nothing else if we pair it with the default cell: its config is difficulty_level: 0.3 — all other parameters default (2–5 km). Its matched d=0 counterpart is therefore cell id_bias30_60 (empty config $\equiv$ defaults). Both cells are therefore the same scenario distribution, differing **only** in the target's evasion.

Three things follow:

- (1) **The direction replicates under an independent method.** Here the maneuver is frozen and only the fire policy varies; in \S4.4/\S4.6 the fire head is re-trained and the maneuver heads are frozen. Two different identification strategies, same answer: **evasion punishes the conservative launch policy and not the aggressive one.** asap rises slightly (93.3 $\rightarrow$ 96.7%) while the inherited policy falls (53.3 $\rightarrow$ 46.7%), so the gap widens by 10.0 pp — the same sign as the +4.25 pp widening measured by the within-policy intervention.
- (2) **The opportunity-set asymmetry is a structural constant, not a difficulty effect.** The legal-window median is 4.0 for asap and 42.0 for the abstaining policy — *identical at both difficulties*. Firing collapses your own window (cooldown removes legality); abstaining inflates it 10 $\times$. This is the measured basis for the \S4.2 caveat, and it is stable across the difficulty change.
- (3) **Cell and difficulty were separated, not conflated.** An earlier reading of this run compared it against the dist2_3k table in \S4.3 and would have attributed part of the difference to evasion when it was actually the 2–3 km range restriction. The matched pair above removes that confound.

4.4 b Figure 2 — the mechanism, made visible

Figure 2 (results/shoot_eval/mechanism_seed20007_d00.png) shows one matched episode (seed 20007, d=0) played by BC and by SPC, with everything else held identical.

- **Panel A** plots the state the decision is taken on — range to target and angle-off (ATA) — with the environment-legal launch windows shaded, every legal step BC declines marked, and both policies' launches marked.
- **Panel B** plots the decisions themselves as three rows: launch mask, BC fire command, SPC fire command.

What the figure is for. It is an *attribution* device, not a trajectory showcase. BC uses **3 of 41** environment-legal steps on this seed (CLR 7.32%, close to the 7.26% aggregate) and times out after 1500 steps; SPC uses **4 of 4** and kills the target at step 842. The state trace is **bit-identical** over all 842 common steps (maximum position

Arm	Kill rate	Wilson 95%	Launches/ep	CLR	lost	hit
Rule expert (fresh-env path, E3)	147/400 = 36.75%	[32.17, 41.58]	2.83	—	0	0.9973
BC round1 (frozen)	186/400 = 46.50%	[41.67, 51.40]	3.06	7.26%	0	0.9984
SPC (this paper)	363/400 = 90.75%	[87.51, 93.21]	3.90	100.00%	0	1.0000

	count
both kill	186
neither kill	37
only SPC kills	177
only BC kills	0
discordant total	177

deviation **0.0e+00 m**) — the maneuver heads are frozen — so the figure contains a *single* state trajectory and the entire difference is confined to Panel B. That is the claim: same geometry, same maneuver, different launch decision.

Two caveats the caption must carry: (i) this is a **single illustrative episode**, chosen to make the mechanism legible — the aggregate evidence is Table 3 (\S4.3) and Table 4 (\S4.5b); (ii) SPC's legal-step count (4) is smaller than BC's (41) because firing triggers the launch cooldown, which removes legality — the policy-dependent denominator documented in \S4.2(b), not a defect of the figure.

4.5 C3(b) — SPC intervention result (primary endpoint, E7)

Setting (all arms identical): $d=0$, deterministic masked argmax, geometry U(0,60), **seeds 20000–20399 (n=400)**, one code path (eval_bc_1v1.py), identity recorded in each file's run_meta (BC sha256 aad05b45\ldots, SPC sha256 36d79bd9\ldots).

Paired BC vs expert on the same 400 seeds: $\Delta\text{kill} +9.75$ pp, win/lose/tie **63 / 24 / 313**, exact McNemar $p = 3.48\text{e-}05$ ($\chi^2_{cc} p = 4.62\text{e-}05$); 72% of the 87 discordant seeds favour BC. $\Delta\text{lost} = 0.00$, $\Delta\text{launches} = +0.23$. This is the first *significance test* attached to the BC-beats-expert claim — the sealed Phase-1 record had only a bootstrap CI ([+3.8, +11.0] pp, 500 seeds, old geometry).

Expert numbers in this paper are fresh-env measurements. A second evaluation path that reuses one environment across all episodes disagrees with it on identical seeds — 32.25% versus 36.75% at $d=0$, and a 20-seed screen shows different per-seed kill vectors rather than noise. Both paths import the same rule functions at the same commanded speed, so neither the rule nor the flight controller explains the gap; the cause is environment lifecycle, since the reused-env path builds a *single* BaseEnv while every other script evaluated here builds a fresh one per episode. The reused-env figures (32.25%, 25.25% at $d=0.3$) are therefore excluded rather than wrong: they are measured under a different episode-construction protocol, and their artifacts are renamed EXCLUDED_reused_env_expert_* so no table can pick them up by accident. The consistency gate is tests/check_expert_path_consistency.sh; Appendix E.1 lists the affected scripts and artifacts.

Paired contingency and test (scripts/paired_mcnemar.py, exact McNemar, no scipy):

- **Paired difference +44.25 pp;** exact McNemar $p = 1.04\text{e-}53$ (χ^2 with continuity correction $p = 5.97\text{e-}40$).
- **The direction is unanimous: all 177 discordant seeds favour the correction; there is no seed where BC succeeds and SPC fails.** This is stronger than a small p-value — it is a *dominance* statement over the evaluated scenario set, and it is the cleanest single number in the paper.
- $\text{lost_target} = 0$ and $\text{launch_quality.bad} = 0$ in both arms — the corrected policy does **not** buy kills with reckless launches: premium 921 / good 639 / bad 0, i.e. **every** SPC launch still scores "good-or-better" under the expert's own quality function.
- Heading/speed behaviour is bit-identical to BC by construction ($G1 \text{ max diff} < 1\text{e-}9$; $G2$ identical action sequences), so the entire gain is attributable to the launch head alone.

Report it bound to its conditions, never as a bare improvement. Use: "Under a matched $d=0$, deterministic-argmax evaluation on 400 paired seeds, replacing the conservative launch decision with SPC raises the kill rate from 46.50% to 90.75% (+44.25 pp, exact McNemar $p = 1.04\text{e-}53$)." Not: "SPC improves performance by 44 pp." The four qualifiers (matched setting, $d=0$, deterministic argmax, isolated intervention) are load-bearing — they are *why* the attribution is legitimate.

Both former docs-only numbers have now been replaced by fresh-env measurements. The rule-expert ID kill rate is **36.75%** (E3 paired, fresh env, seeds 20000–20399) — *not* the docs' 35.8%, and *not* the reused-env 32.25%. The rule-oracle ID kill rate is **90.75%** (E2), measured on the same seed set as SPC and exactly equal to it; that equality is a completeness check, not independent validation (same policy by construction, \S4.4). The originals (paired_bc_vs_expert_500.json, asap_baseline.json) remain lost — the replacements are new measurements, not recoveries, and were taken under U(0,60) rather than the docs' U(30,60).

SPC vs the rule oracle on one seed set (E2, seeds 20000–20399, n=400). Resolved: eval_asap_baseline.py had no `-start-seed`, so the oracle could only ever run seeds 0. .N-1 and could not be matched to a trained policy's evaluation. Patched, then run on the same seed set.

Identical on every reported metric. Identities verified from run_meta: the oracle arm loaded sha256 aad05b45\ldots (BC round1 — the *frozen maneuver* source, correct), SPC loaded 36d79bd9\ldots.

This is a completeness check, not independent validation. The two are the *same policy by construction*: the oracle is "frozen BC maneuver + fire on every legal step", and SPC is "bit-identical frozen maneuver heads + a launch head whose measured CLR is 100.00%", which also fires on every legal step. Their agreement is therefore expected rather than evidence that two independent methods concur. What it does establish is that SPC captures the **entire** oracle gap — 46.50% $\rightarrow$ 90.75%, i.e. 100% of it — rather than some fraction. The independent replication in this paper is \S4.3(E6)

Arm	Kill	Launches/ep	Hit	lost	Termination reasons
Rule oracle (asap: frozen BC maneuver + fire on every legal step)	363/400 = 90.75%	3.90	1.0000	0	target_killed: 363, timeout: 37
SPC	363/400 = 90.75%	3.90	1.0000	0	target_killed: 363, timeout: 37

Comparison (d=0, new geometry U(0,60))	n	Kill rate	Paired diff	McNemar
SPC base @ U(0,60)	400	362/400 = 90.50% (Wilson [0.872, 0.930])	—	—
SPC retrained under U(0,60)	400	375/400 = 93.75% (Wilson [0.909, 0.957])	+3.25 pp	p = 0.0146
same pair, unpaired Fisher	400	—	—	p = 0.1146 (n.s.)

against \S4.4/\S4.6, where the identification strategies differ in kind (off-policy enumeration versus within-policy intervention).

One limitation: `eval_asap_baseline.py` writes aggregates only, with no per-episode detail, so the identity above follows from construction rather than from a seed-by-seed match. Emitting per-episode records would make it mechanically checkable.

4.6 Robustness check 1 — widened initial geometry (heading bias) — a benchmark check, not a contribution

Compressed to one subsection per scope decision: **this is a benchmark-change + retraining robustness check, NOT a second contribution.**

- **Same weights, geometry-only change** (n=100, s42): 87% → 91% ⇒ widening the bias range makes the benchmark easier by $\approx +4.5$ pp. This is a *benchmark* effect and must be reported as such.
- The +3.25 pp retraining gain is real and paired-significant (19 vs 6 discordant pairs, 76% favouring the retrained arm), robust across 4/4 bias bins (+1/+5/+5/+2 pp).
- **Caveats to state in the paper:** single seed family (seeds 42–441); d=0 only; Bonferroni $\alpha=0.0125$ would put $p=0.0146$ just outside; JSBSim physics is not cross-machine reproducible.
- **Framing rule:** never present 87→95 as the effect of the launch-head correction. The launch-head correction is \S4.4.

4.7 Robustness check 2 — target evasion (difficulty_level = 0.3)

Target adds S-turn $\pm 30^\circ \cdot d \cdot \sin(0.3t)$ plus a missile-threat break-turn and a dive to $-800 \cdot d$ m (floor 2000 m). Identical protocol to \S4.4 — d=0 vs d=0.3 differ **only** in -difficulty, same seeds 20000–20399, same code path, identity recorded in each file’s `run_meta`.

Table 4 — main results: kill rate and paired tests. Fresh env per episode, seeds 20000–20399, geometry U(0,60), deterministic masked argmax; difficulty is the only variable between the two columns’ groups. CLR values live in Table 2 (\S4.2) and are not repeated here. Every row is generated by `scripts/collect_matrix.py` from each artifact’s own `run_meta` — not assembled by hand.

Read against Table 2’s CLR column, the table says three things at once — the teacher is restrictive, imitation transmits it, and evasion punishes it in proportion to how restrictive the policy is (**expert -11.00 > BC -4.25 > SPC 0.00**).

Paired tests, both against the same 400 seeds (primary protocol: d=0 and d=0.3, geometry U(0,60), deterministic masked argmax, fresh env per episode). The expert-vs-BC comparison and the SPC-vs-BC comparison share a seed set, so both are paired by construction rather than by assumption:

Per-seed flips under evasion: expert **+29 gained** / **-73 lost**, BC **+25** / **-42**. Both are loss-dominated, but the expert’s net is -44 seeds against BC’s -17 — the asymmetry, not just the mean, drives the ordering above.

Five findings:

- (1) **The correction is robust, and its benefit grows under evasion.** SPC’s kill rate is *unchanged* (90.75% at both difficulties) while BC loses 4.25 pp, so the SPC-BC gap widens from +44.25 to **+48.50 pp**, and the discordant count rises from 177 to 194. **This widening is independently replicated by a different identification strategy:** in the frozen-trajectory oracle enumeration (\S4.3, E6) the asap-inherited gap widens from **+40.0 to +50.0 pp** across the same difficulty change. One method re-trains the launch head with the maneuver frozen; the other freezes the maneuver and swaps in rule-based launch policies. Both say the same thing: evasion punishes the restrictive launch policy, not the permissive one.
- (2) **The restrictive preference is structural, not scenario-specific — and measured on both policies.** BC’s CLR barely moves (7.26% → **6.95%**), and the expert’s moves just as little (**6.07%** → **5.40%**): the designed gate suppresses launches to the same degree whether or not the target manoeuvres. The teacher’s restriction is therefore not caution that pays off when the target turns — it is a fixed property of the predicate, and the imitation inherits it as one too.
- (3) **The identical aggregate is not an artefact — the difficulty change was verified to take effect.** 363 kills at both difficulties admits a trivial explanation, so it was tested directly: **368/400 episodes change length** (so difficulty=0.3 is applied) and **382/400 seeds keep the same kill outcome, with 18 flips split 9 gained / 9 lost**. The match is a genuine near-cancellation rather than a no-op.
- (4) **Evasion cost is ordered by how restrictive the launch policy is.** Expert **-11.00 pp**, BC **-4.25 pp**, SPC **0.00 pp**, all three legs on fresh-env evaluations over the same seeds. The per-seed flips indicate the mechanism: the expert loses 73 seeds and gains 29, while BC loses 42 and gains 25 — a suppressed launch window is unrecoverable once the target

Policy	d=0 kill	d=0.3 kill	Δ (evasion cost)
Rule expert (fresh env)	36.75% (147/400)	25.75% (103/400)	-11.00 pp
BC round1 (inherited launch policy)	46.50%	42.25% (169/400)	-4.25 pp
SPC (corrected launch head)	90.75%	90.75% (363/400)	0.00

Paired comparison	d=0	d=0.3	Test
BC - expert	+9.75 pp (63/24/313, discordant 87, 72% favouring BC)	+16.50 pp (85/19/296, discordant 104, 82% favouring BC)	exact McNemar 3.48e-05 / 3.79e-11
SPC - BC	+44.25 pp (discordant 177 : 0)	+48.50 pp (discordant 194 : 0)	exact McNemar 1.04e-53 / 7.97e-59

Intervention	Result (400 seeds, d=0)	What it tests
A6 — full-network fine-tune (everything trainable)	77.50% kill, -13.25 pp vs SPC (exact McNemar $p = 1.33e-11$), maneuver logits move by 8.51	is <i>freezing</i> merely an engineering convenience? → no: it is the identification
A5 — random-init launch head, encoder frozen	90.75% kill, 0 discordant seeds vs SPC , maneuver deviation exactly 0.00e+00	does the correction depend on warm-starting from BC's launch head? → no

Identity measurement	SPC (fire head only)	A6 (full network)	Reading
heading/speed logit max-diff vs θ_{BC}	< 1e-9	8.51	the maneuver is provably untouched under SPC and provably <i>changed</i> under A6
heading/speed action sequences identical	yes	no	same conclusion, measured behaviourally rather than in logit space
launch-mask agreement	yes	yes	both reach the target behaviour on the launch head
validation allowed-window accuracy	≥ 0.995	0.9967	neither arm can be dismissed as under-trained

turns away. The ordering rests on a single seed family, so it is reported as a well-supported ordering rather than a law.

- (5) **The imitation gap itself widens under evasion.** BC's paired margin over the expert goes from **+9.75 pp** (d=0) to **+16.50 pp** (d=0.3), i.e. it more than doubles. This is a new observation and a useful one: the value of imitating-then-correcting this expert is *larger* in the harder regime, which is the opposite of the usual expectation that a stronger expert is needed as the problem gets harder.

4.8 Ablations: why *surgical*?

Two ablations, both at the paper's primary protocol (400 seeds, d=0, Table 4), ask whether the locality of the correction is doing real work or is an arbitrary design choice. The results indicate that **locality is the key factor enabling an attributable correction:**

A6 — full-network fine-tuning on the same objective. Identical objective, data, loss and initialisation as SPC; the only change is that every parameter is trainable rather than the launch head alone, at a deliberately **more generous** budget (–epochs 40 –lr 3e-4 versus –epochs 15 –lr 1e-2), so that a negative result cannot be attributed to under-training.

A6 result — the maneuver moves, and the correction is still worse than doing nothing clever. Both pre-registered measurements came back on 2026-09-18.

(i) *Maneuver deviation.* Identical objective, identical data, identical loss, identical initialisation; the only change is that every parameter is trainable rather than the launch head alone, at a **more**

generous budget (–epochs 40 –lr 3e-4 versus –epochs 15 –lr 1e-2) so the ablation cannot be dismissed as under-trained.

(ii) *Kill rate and CLR under the paper's primary protocol* — d=0, U(0,60), the same 400 seeds as Table 4, so this is directly comparable rather than a screening number:

The reading is that freezing is doing real work, on both axes at once. Full-network fine-tuning recovers most of the gap (46.50% → 77.50%) but remains **significantly worse than the localized intervention on identical seeds** (+13.25 pp in SPC's favour, $p = 1.33e-11$, 60 of 67 discordant seeds favouring SPC), and it pays for that with an 8.51-logit change to the maneuver — it is less effective **and** no longer attributable. These results indicate that allowing unrestricted parameter updates weakens attribution and does not recover the performance achieved by the localized intervention.

A5 (the control). Keep SPC's freeze pattern exactly — encoder and every maneuver head frozen at θ_{BC} — and rebuild *only* the fire head from `nn.Linear.reset_parameters()`, the constructor's own initialisation, so the sole difference from SPC is where the moving head starts. Trained on the same 300-episode rollout set, the random head reaches **100% allowed-window recall in its first epoch** and early-stops, with the identity gates reporting **exactly zero** maneuver deviation. At the primary protocol it scores **90.75%** (363/400), and the paired test against SPC reports **zero discordant seeds**: both arms kill the same 363 seeds. Training detail and the 60-seed screening comparison are in Appendix E.

What zero discordance does and does not license. It is *not* an equivalence test: McNemar with no discordant cells returns p

Arm (400 seeds, d=0)	Kill rate	CLR	vs SPC (paired)
BC (inherited launch policy)	46.50%	7.26%	—
A6 — full-network fine-tune	77.50% (310/400)	99.67%	-13.25 pp , discordant 67 (60 SPC / 7 A6), exact McNemar p = 1.33e-11
SPC — launch head only	90.75% (363/400)	100.00%	—

Arm (60 seeds, same seed set)	id cell	dist2_3k cell	window utilisation (id)
BC (inherited launch policy)	40.0%	15.0%	7.2%
A6 — full-network fine-tune	81.7%	66.7%	98.3%
mask-permissive rule oracle	95.0%	86.7%	100%

A7 CLR for other binary decisions conceptual generality of the metric (future work)

= 1.0, the absence of a detected difference rather than a proof of equality, and no TOST is run here. The supportable statement is that A5 and SPC are **behaviourally indistinguishable under the evaluated metric** — on 400 seeds and this benchmark the per-seed kill vector is identical, the only trace of the different initialisation being one extra legal step (1561 vs 1560). We do not claim *identical* or *equivalent* policies; no action-logit comparison is made, and a stronger equivalence claim would require a pre-registered margin.

What the pair establishes. Unfreezing everything *hurts* significantly and destroys the attribution; re-initialising the one head that is supposed to move makes no measurable difference at all. The observed benefit is therefore attributable to **the choice of editable parameters** — which decision dimension is allowed to change — rather than to the amount of trainable parameters or the initialisation of the launch head.

A6b — the screening set, kept because it carries a third arm. The ablation script's own 60-seed evaluation (geometry U(0,60), deterministic masked argmax) runs the id and dist2_3k cells side by side and adds the **mask-permissive rule oracle**, which the 400-seed run above does not:

A6 also loses to a rule that simply fires whenever the mask permits (81.7% vs 95.0%). The screening set is **60 seeds** and therefore carries roughly ± 10 pp of small-sample spread — the same seeds give BC 40.0% where n=400 gives 46.50% — so these figures are quoted only for the third arm and never alongside the 400-seed numbers as one comparison.

5 DISCUSSION + CONCLUSION (0.5 PAGE)

5.1 Why this is a general concern (and why it is an *interactive-systems* concern)

- Rule-based experts are **designed**, not optimized; a locally plausible quality gate can be globally costly in a closed loop, and the cost is invisible in the expert's own statistics.
- BC is a *faithful* propagator: the bias becomes a floor under every downstream method and every evaluation.
- The failure mode is characteristic of interacting-agent environments: the "correct" launch time depends on the trajectory the launch itself induces, so a static quality predicate cannot be validated offline.

- CLR is cheap, needs no oracle, and applies to any policy with a discrete commit-or-not decision (launch, hand-off, emergency stop, lane change).

5.2 Limitations

- **1v1 WVR only**; N-vs-N extension is future work (and the thesis's core).
- **Single expert, single designed gate**: we identify one instance of the failure mode, not its prevalence.
- **Suboptimality is established conditionally**: on the frozen BC maneuver trajectory (\S4.3) and within the frozen-heads intervention (\S4.4). We do not claim the mask-permissive policy is optimal in general.
- **Statistical scope**: the +3.25 pp geometry result is a single seed family and fails a Bonferroni $\alpha=0.0125$.
- **Evasion is a single interpolated setting** (difficulty = 0.3, one scripted evasion family), not a learned or optimised opponent. "Interactive" here means a reacting scripted target, not an adversary trained against the policy.
- SPC requires knowing *which* head to correct; automating that (e.g. per-head CLR + ablation ranking) is open.

5.3 Future Work

- Automate head selection from per-head diagnostics.
- N-vs-N: per-agent CLR and per-agent surgical correction; does the bias compound with fleet size?
- Launch *timing* when launch cost/ammo scarcity makes launching on every mask-permitted step genuinely suboptimal — the regime where PPO becomes meaningful (project record documents that the current reward admits no such trade-off).
- Integrate into the hierarchical tactical architecture planned for the thesis.

5.4 Claim statement (exact wording to use)

The sentence the paper exists to support (2026-09-18): *the failure was localized to a single discrete decision dimension, and correcting only that dimension recovered oracle-level performance while preserving the original maneuver policy.* Everything below is scoping, not hedging — this sentence is the claim, the qualifiers are what keep it honest.

We do **not** claim that the mask-permissive policy is optimal in general. We claim that **a behaviorally isolated intervention reveals the inherited launch policy to be suboptimal in the studied regime**: with the maneuver policy provably frozen ($\max \text{logit_diff} < 1e-9$, identical action sequences), correcting the launch head alone raises the kill rate from 46.50% to 90.75% on 400

paired seeds (+44.25 pp, exact McNemar $p = 1.04e-53$, with all 177 discordant seeds favouring the correction and none favouring the original), and off-policy enumeration on a fixed maneuver trajectory shows the inherited launch policy to be dominated by the mask-permissive policy within the enumerated fire-policy family.